

This emerging vertical opens up several business opportunities for technopreneurs. One of the most obvious options is to set up a tech-enabled vertical farm, grow stuff and sell it, like Aerofarms and Spread. Other options include designing and implementing tech solutions for emerging vertical farms, or selling individual technologies for the same.

Off-the-shelf products for individuals and farms. Developing a vertical farming solution need not necessarily be complex. Look at what Bengaluru based startup Greenopia has done. They sell kits with smart self-watering pots, enriched soil and the right seeds. The sensor-embedded pots replenish moisture in the soil on a need basis, and notify you when you need to refill water externally (once in a few days).

The solution comes with a mobile app, which not only helps you be in control of your plants—be it flowers or herbs—but also share information with a growing community. For example, if somebody has managed to consistently grow the perfect mint in their kitchen garden, they can share the parameters that led to the perfect crop, so others can replicate it.

You could start with a simple product and, as your business gains traction, you could expand your portfolio with more products. One well-known example of this model is Wyoming (USA) based Bright AgroTech. They sell a variety of technology products ranging from vertical growing towers and LED lights to software. They also offer consultation, which is very important for a nascent vertical like this.

Focusing on one of the supporting technologies. You can develop and sell solutions focused on any one aspect of vertical farming, like air-conditioning, lighting or intelligence. However, you need to think carefully before entering into a

Known benefits of vertical farming

Predictable. Total control over environment ensures reliable and high yields, unaffected by the vagaries of nature.

Profitable. Although investment and cost appear to be high, experts feel that this is offset by reduced labour requirement and absence of equipment like tractors, seed dispersers, sprinklers and other farming equipment.

Returns are also assured. Aerofarms, the world's largest indoor vertical farm, claims that given the same seed, they can grow it in half the time as a traditional field farmer, and achieve 75 times more productivity per square foot than a commercial field farm.

Safe. Indoor farming is clean and hygienic. The sterile environment keeps pests away, and also keeps unclean farming practices at bay.

Eco-friendly. Although the carbon footprint of the tech gadgetry appears high, evangelists of vertical farming say that it is compensated by reduction in transport and pollution, economical use of water, greater efficiency in usage of land and so on.

Healthy. Some doubt whether the vegetables grown indoors, devoid of natural sunshine, will be as nutritious as natural field-grown vegetables. But, Caleb Harper who drives CityFARM project of MIT Media Lab, explains that, there is absolutely no nutritional difference between plants grown in sunlight and under an LED because plants only harvest ten per cent of the sun's rays, which can be recreated in the lab.

Socially beneficial. Like in the case of Vertical Harvest, a social organisation based in Jackson, Wyoming, USA, vertical farming can be a source of livelihood for physically-challenged people and other less-privileged individuals.

business that covers only one tech component, because your success might depend on the availability of other supporting technologies in that region.

In most cases, you might have to supply your tech to a larger player, who offers an end-to-end platform. That is, you become a vendor. Plus, in an emerging market like India, you cannot expect a large enough market to sustain your business if you plan to sell only to vertical farmers. This option makes sense if vertical farming is a logical extension to an already existing product line; like when Philips or Illumitex forayed into horticultural lighting solutions.

Platform approach. A more holistic business would be to aggregate all the technologies required for vertical farming onto one platform. To achieve this, you would have to put together a multi-disciplinary team to design and implement the solution. When doing so, it is clear that you cannot succeed if you focus only on technology aspects. You need to pay more attention to farming aspects

and use technology as it is meant to be—merely an enabler.

Dr Dickson Despommier, author of *The Vertical Farm*, hits the nail on the head, when he says, "Chances are, none of the fanciful, futuristic Jetsons vertical farms that you will find on Google Images [will] be built, as they are not considering the reality of farming on a commercial scale. Many of these designs are proposed by architects and designers, not the farmers themselves."

So if you want to succeed, ensure that your team has a farmer or two. This is a good business model, which has worked for companies like Houston (USA) based Indoor Harvest Corp. They partner with companies like Illumitex, Dosatron and Hort Americas to work out all aspects of a vertical farm, right from watering, nutrient delivery, lighting and HVAC, to constructing expandable and upgradable modular infrastructure to house it all.

Technology as a service. Vertical farming need not always be on a commercial scale, conducted